SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

Annexure A

<u>Analytical Problem Solving</u>

Code of Conduct:

I K. Reshma (192565068) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Reshma

Level ->

Student 22: Kadapa Reshma | Register Number: 192565062

1. Consider the map of Romania (Arad to Bucharest) as a graph of cities connected by roads with distances. Trace the order in which cities are expanded using Breadth-First Search and then Depth-First Search, starting from Arad. Compare the two traces in terms of completeness and optimality.

2. For a Tic-Tac-Toe playing agent, define its PEAS description and characterize the environment (is it fully observable? deterministic? episodic? single or multi-agent?). Explain why this environment is classified as adversarial.

3. A smart thermostat adjusts room temperature based on current readings and a target temperature. Design it first as a Simple Reflex Agent, then redesign it as a Model-Based Reflex Agent. Explain what internal state the model-based version must maintain and why it performs better.

Romanea Map problem (Arab to Bucharest)

Arekfieal Intellegence uses search algorethms to solve path finding problems efficiently. The Romanea map is a classical example used to demonstrate uninformed search techneques such as Breadth-first search and Depth Ferst search. These Algorethms detter in the order of node expansion, memory requerements, complete nessand optemidety.

Breath First search

BFS explores all neighboring nodes before moving to the next level.

Expanding order:

* Arad
* Zerend
* sebeu
* Timesoara
* oradea.
* Fagaras
* Rimnecu velcea
* Lugoj
* Bucharest

Solution path:-

Arad → Sebeu → Fagaras → Bucharest.

Depth First Search

DFS explores one branch as deeply as possible before backtracking.

Expanding order:-

* Arad

* Zerend

* Oradea

* Sebeu

* Fagorus

* Bucharest

Solution path:-

Arad → Zerend → Oradea → Sebeu → Fagorus → Bucharest.

Comparison of BFS and DFS

* BFS searches level by level DFS searches depth-wise.

* BFS gurantees finding a solution if one exists; DFS may fail in infinite search space.

* BFS always finds the shortest path in an unweighted graph, making it optimal.
* DFS is not optimal because it may find a longer path first.
* BFS requires more memory due to storing all frontier nodes.
* DFS requires less memory and is simpler to implement.

Conclusion
BFS is prefered when the shortest path is required, while DFS is suitable when memory is limited and quick exploration is desired.

(2) Tic - Tac - Toe playing Agent

Tic Tac Toe is a popular game in Artificial Intelligence used to study intelligent agent and adversarial search. An AI agent must observe the board, make decision, and respond to an opponent's actions.

PEAS description
* performance Measure :-

* environment.
* Actuators
* sensors.

Environment characterestics

1. **Fully observable**
   * The entire board is vesible to both players at all times.
   * No hedden information exists.
2. **Determenestic**
   * Every action produces a predectable result.
   * there is no randomness involved.
3. **sequental**
   * Each move affects all future moves.
   * The outcome depends on the sequence of action.
4. **static**
   The board remain unchanged while a player is making a deceseon.
5. **Multi Agent**
   Two agents (players) interact in the environment.

Why Tic-Tac-toe is Adversarial

* The game involves competition b/w two players.
* One player's objective is to defeat the other.
* Each player attempts to block the opponent's winning strategy.
* The utility gained by one player result in a loss for the other.
* Therefore, Tic-Tac-Toe is classified as an adversarial environment.

Conclusion

Tic-Tac-Toe demonstrates how intelligent agents operate in competitive environments and form the basis for understanding advanced game playing algorithm such as Minimax and Alpha Beta pruning.

(3) <u>smart Thermostat Agent</u>

A smart thermostat is an intilligent temperature system that maintains room temperature automatically. It senses environmental conditions and performs action to ensure user comfort and energy efficiency.

<u>Simple Reflex Agent Design</u>

* If temperature is below the target value, switch ON the heater.
* If temperature is above the target value, switch ON the cooler.
* If temperature equal the target value, maintain the current state.

<u>For example:-</u>

* Target Temperature = 24°c
* current temperature = 20°c
* Action = Heater ON.

<u>lemetations of simple Reflex Agent</u>

* No memory of previous temperatures.
* cannot predict future changes.

May repeatedly switch devices ON and OFF.

* less energy efficient.

Model Based Reflex Agent Design

A Model Based Reflex agent maintains an internal representation of the environment.

It stores:-

* previous temperature readings.
* Heater and cooler status.
* Temperature trends.
* Target temperature settings.
* User preference.
* Time of day.

Working of Model - Based Agent

* continuously monitors temperature changes.
* Determines whether the room is heating or cooling.
* predicts future temperature conditions.
* Makes intelligent decisions to maintain stability.

For example:

* previous Tempurature = 22°c
* Current Temperature = 23°c
* Target Temperature = 24°c.
* Decision: continue heating briefly instead of switching repeatedly.

Advantages of Model Agent

* Maintaes an internal state.
* provides stable temperature regulation.
* Reduces powers consumption.
* Improves user comfort.
* Makes better decisions using historical data.

Conclusion:-

The Model - Based Reflex Agent outperforms the simple Reflex Agent because it uses memory and enviromental modeling to make intelligent decesions.